

**Assignment
on
EQ-5D**

**Course
Econ 507 - Health Economics**

Submitted to
Kawsar Hossain Sojib
Lecturer, Department of Economics, Jahangirnagar University

Submitted by

Name	Roll
Syed Mahin	498
Muhammad Shakib Hossain	500
Abdullah Bin Ashraf	521
Samia Jahan	531
Raiyan Rahman Rabi	532
Mst. Nazmin Akter	549
Mst. Farhana Khatun	551
Shaloya Zabin	550
Rubaiya Zahan Mim	538
Nafis Ferdous Sakib	1818

Submission on: 20 June 2023

EQ-5D

1. Definition

The EQ-5D is a well-established and widely-used generic instrument for assessing health-related quality of life. EQ-5D is a standardized measure of health-related quality of life developed by the five-country group (Finland, the Netherlands, Norway, Sweden and the UK) that developed the EuroQol instrument over the period 1987–1991 to provide a simple, generic questionnaire for clinical and economic appraisal and population health surveys. The questionnaire was renamed ‘EQ-5D’ in 1995.

EQ-5D assesses health status in terms of five dimensions of health and is considered a ‘generic’ questionnaire because these dimensions are not specific to any one patient group or health condition. EQ-5D can also be referred to as a patient-reported outcome (PRO) measure because patients can complete the questionnaire themselves to provide information about their current health status and how these changes over time. ‘EQ-5D’ is not an abbreviation and is the correct term for the instrument in general.

The EuroQol five-dimensions (EQ5D) is a versatile quality of life (QOL) instrument with five dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression) and a visual analogue scale. It can be used to calculate quality-adjusted life years.

The EQ-5D comprises five questions on mobility, selfcare, pain, usual activities, and psychological status with three, or five possible answers for each item. A summary index with a maximum score of 1 can be derived from these five dimensions by conversion with a table of scores. The maximum score of 1 indicates the best health state, by contrast with the scores of individual questions, where higher scores indicate more severe or frequent problems. In addition, there is a visual analogue scale (VAS) to indicate the general health status with 100 indicating the best health status.

2. Purpose / Use

EQ-5D is widely used around the world in clinical trials and real-world clinical settings, population studies, health economic evaluations and many other types of studies where a generic measure of health status is useful. By mid-2020, the number of EQ-5D studies registered with the EuroQol Group totalled over 39,000. These comprised over 80 clinical areas and related to surgical procedures, hospital waiting lists, physiotherapy, general practice and primary care, and rehabilitation. The number of annual requests to use EQ-5D is approximately 5000, and EQ-5D data have been reported in over 8000 peer-reviewed papers over the past 30 years.

EQ-5D can be used for a variety of purposes.

In clinical trials:

In clinical trials and routine clinical settings, EQ-5D can be used

- (i) to provide a profile of patient health on the day of questionnaire completion;
- (ii) to monitor the health status of patient groups at particular times, e.g. at referral, admission, discharge, and follow-up;
- (iii) to measure changes in health status over time in individual patients and in cohorts of patients, such as before and after health interventions and treatments.

In population health surveys:

Data collected with EQ-5D can be used to assess and compare health status between groups of patients, between patients and the general population, or between the general populations of different countries.

In medical decision-making:

EQ-5D can be used

- i. to measure the impacts and outcomes of healthcare services;
- ii. to provide relevant information for the economic evaluation of health programmes and policies;
- iii. to assist in providing evidence about effectiveness in processes where drugs or procedures require approval.

In economic evaluation:

EQ-5D is the most widely used health-related quality of life questionnaire in health economic evaluations. EQ-5D can be used to derive a set of values that reflect people's opinions of the relative importance of different health problems. These values can be used to derive QALYs for application in cost-effectiveness and cost-utility evaluations.

Routine outcome measurement:

Longitudinal data collected with EQ-5D can be used at the individual level to monitor a patient's health status over time, at the institutional (e.g. hospital) level to monitor the institute's performance as a healthcare provider, or at the national level to monitor the population's health over time.

Actually EQ-5D is used by academia (e.g. universities, research institutes), governments (national, regional, municipal), industry (e.g. pharmaceuticals, medical devices) and by hospitals and clinics.

At an international level, the pharmaceutical industry and scientific groups are the foremost users of the instrument. EQ-5D is frequently used in (multinational) clinical trials by pharmaceutical companies. As EQ-5D is available in many languages in a standardized format, it can potentially be used to explore differences in outcome across borders. With translations available for more than 170 countries/regions, EQ-5D is used globally.

At a national level, the institutions involved are country/region-specific and use of EQ-5D in the supply and demand areas of health care is widespread. The following list gives some idea of the personnel involved:

- General practitioners
- Hospital specialists
- Nurses and remedial therapists
- Researchers into clinical practice
- Those responsible for audit and quality assurance
- Public health specialists
- Health service researchers
- Managers in health care institutions

3. Requirements

EQ-5D was designed as a self-completed questionnaire to fulfill two functions: (i) to provide a descriptive profile of current health status; and (ii) to provide a way of assigning a single numerical value to each of the possible health states described by the descriptive system, for use in economic evaluations of health care.

The requirements for its design were that...

- The dimensions should be relevant to both patients and members of the general population;
- The descriptive system should be simple – with as few dimensions as possible, and as few levels as possible within each dimension;
- It should be short and easily self-completed in a range of settings, and simple enough not to require detailed instructions; and
- It should be reliable, valid, and able to identify changes in health status related to illness and health interventions.

EQ-5D Registration

The EuroQol Research Foundation is a registered non-profit organization in the Netherlands. The Foundation serves as the single point of distribution of the family of EQ-5D instruments.

No license fee will be charged if the EQ 5D is used for a non-commercial purpose. For most requests, one will only need to agree to the Terms of Use in order to obtain EQ-5D. However, in some cases a license agreement is needed, for instance when the sample size is higher than 10,000. If anyone is planning to collect data digitally on a non-supported platform one will be asked to sign a Digital Use Letter.

A license fee will be charged for commercial use of the EQ-5D. For example, when EQ-5D is used by or on behalf of a pharmaceutical company, medical device manufacture, or any other for profit company a license fee will be charged.

EQ-5D, Digital Data Collection

From 6 April 2022 onwards, a new digital format for EQ-5D has been made available. This new digital format can be used on any digital device and will be made available for all EQ-5D versions (3L ,5L and Y) in all languages that are currently available digitally. As such, these versions can no longer be selected via the registration pages on the EuroQol website. Customers with an open terms of use agreement, or an open license agreement, will keep access to these “legacy” EQ-5D formats. As a courtesy, these existing customers will be given access to the new digital format of their registered legacy EQ-5D versions.

4. Constructive Method

The questionnaire ‘EQ-5D’ comprises a family of questionnaires: the three-level EQ-5D-3L, the five-level EQ-5D-5L, and the youth version EQ-5D-Y. The EQ-5D essentially consists of two pages: the EQ-5D descriptive system and the EQ-5D visual analogue scale (EQ VAS).

Descriptive Systems

The EQ-5D descriptive system comprises five dimensions: mobility, self-care, usual activities, pain and discomfort, and anxiety and depression. The number of levels in these dimensions differ in the EQ-5D-3L (three levels) and the EQ-5D-5L (five levels). The EQ-5D-Y has the same five dimensions, but they are worded more appropriately for young people.

EQ-5D-3L

In EQ-5D-3L, the five dimensions each have three response levels of severity. The mobility dimension ranges from 'I have no problems walking about' to 'I am confined to bed'. The self-care dimension ranges from 'I have no problems with self-care' to 'I am unable to wash or dress myself'. The usual activities dimension concerns work, study, housework, family, or leisure activities and ranges from 'I have no problems doing my usual activities' to 'I am unable to do my usual activities'. The pain/discomfort dimension ranges from 'I have no pain or discomfort' to 'I have extreme pain or discomfort'. The anxiety/depression dimension ranges from 'I am not anxious or depressed' to 'I am extremely anxious or depressed'.

Respondents are asked to choose the statement in each dimension that best describes their health status on the day they are surveyed. Their responses are coded as a number (1, 2, or 3) that corresponds to the respective level of severity: 1 indicates no problems, 2 some problems, and 3 extreme problems. In this way, a person's health state profile can be defined by a 5-digit number, ranging from 11111 (having no problems in any of the dimensions) to 33333 (having extreme problems in all the dimensions). The health state 12321 would indicate a person having no problems with mobility, having some problems with self-care, being unable to perform their usual activities, having some pain or discomfort, and not being anxious or depressed. In total, the EQ-5D-3L describes 243 ($=3^5$) potential health states.

Figure 1 / UK (English) EQ-5D-3L Paper Self-Complete (sample version)

Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY

- I have no problems in walking about ☐
- I have some problems in walking about ☐
- I am confined to bed ☐

SELF-CARE

- I have no problems with self-care ☐
- I have some problems washing or dressing myself ☐
- I am unable to wash or dress myself ☐

USUAL ACTIVITIES (e.g. work, study, housework, family or leisure activities)

- I have no problems with performing my usual activities ☐
- I have some problems with performing my usual activities ☐
- I am unable to perform my usual activities ☐

PAIN / DISCOMFORT

- I have no pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have extreme pain or discomfort ☐

ANXIETY / DEPRESSION

- I am not anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am extremely anxious or depressed ☐

7

EQ-5D-5L

EQ-5D-5L was introduced in 2009 with the aim of enhancing instrument sensitivity and providing respondents with the opportunity to provide a more detailed and accurate picture of their health. The EQ-5D-5L descriptive system uses the same five dimensions as the EQ-5D-3L but has two extra levels of severity in each dimension. The five levels in each dimension are worded as (1) 'not/no problems', (2) 'slight problems', (3) 'moderate problems', (4) 'severe problems', and (5) 'unable to' (mobility, self-care, usual activities), 'extreme' (pain/depression), or 'extremely' (anxiety/depression). A few changes in the wording of some levels were also made. For example, in the mobility dimension, the 3L term 'confined to bed' has been replaced with 'unable to walk about' in the 5L; and the first level of self-care in the 5L now refers to washing and dressing, to be consistent with the other levels. Because of the additional levels in EQ-5D-5L, the descriptive system describes 3125 ($=5^5$) potential health states.

Making any EQ-5D (sample) version available on a publicly accessible webpage is **not allowed**. For reproduction/ displaying any EQ-5D sample version, please submit a request for permission by using the EQ-5D registration form.

Figure 1/UK (English) EQ-5D-5L Paper Self-Complete (sample version)

Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY

- I have no problems in walking about ☐
- I have slight problems in walking about ☐
- I have moderate problems in walking about ☐
- I have severe problems in walking about ☐
- I am unable to walk about ☐

SELF-CARE

- I have no problems washing or dressing myself ☐
- I have slight problems washing or dressing myself ☐
- I have moderate problems washing or dressing myself ☐
- I have severe problems washing or dressing myself ☐
- I am unable to wash or dress myself ☐

USUAL ACTIVITIES (e.g. work, study, housework, family or leisure activities)

- I have no problems doing my usual activities ☐
- I have slight problems doing my usual activities ☐
- I have moderate problems doing my usual activities ☐
- I have severe problems doing my usual activities ☐
- I am unable to do my usual activities ☐

PAIN / DISCOMFORT

- I have no pain or discomfort ☐
- I have slight pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have severe pain or discomfort ☐
- I have extreme pain or discomfort ☐

ANXIETY / DEPRESSION

- I am not anxious or depressed ☐
- I am slightly anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am severely anxious or depressed ☐
- I am extremely anxious or depressed ☐

© EuroQol Research Foundation. EQ-5D™ is a trade mark of the EuroQol Research Foundation

EQ-5D-Y

The EQ-5D version (EQ-5D-Y) was introduced by the EuroQol Group in 2009 as a more suitable questionnaire for children and adolescents. It is based on the EQ-5D-3L, but the wording has been modified to be more easily understood and relevant for younger people. The dimensions are: mobility (walking about); looking after myself; doing usual activities (e.g., going to school, hobbies, sports, playing, doing things with family and friends); having pain or discomfort; and feeling worried, sad, or unhappy. The levels for the first four dimensions are: ‘none/no problems’, ‘some (problems)’, and ‘a lot (of problems)’. The levels in the feeling worried, sad, or unhappy dimension are: ‘not’, ‘a bit’, and ‘very’. EQ-5D-Y is recommended for use with 8–11-year-olds and for 12–15-year-olds (although the adult version may be appropriate in the older age group, depending on study design; see Table 1. For children aged 4–7 years, a proxy version should be used – i.e. a version of the questionnaire that is suitable for completion by a third party (e.g. a parent, caregiver, or health professional) on the child's behalf.

Figure 1/UK (English) EQ-5D-Y Paper Self-Complete
(sample version, v2.1)

Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY (*walking about*)

- I have no problems walking about ☐
- I have some problems walking about ☐
- I have a lot of problems walking about ☐

LOOKING AFTER MYSELF

- I have no problems washing or dressing myself ☐
- I have some problems washing or dressing myself ☐
- I have a lot of problems washing or dressing myself ☐

DOING USUAL ACTIVITIES (*for example, going to school, hobbies, sports, playing, doing things with family or friends*)

- I have no problems doing my usual activities ☐
- I have some problems doing my usual activities ☐
- I have a lot of problems doing my usual activities ☐

HAVING PAIN OR DISCOMFORT

- I have no pain or discomfort ☐
- I have some pain or discomfort ☐
- I have a lot of pain or discomfort ☐

FEELING WORRIED, SAD OR UNHAPPY

- I am not worried, sad or unhappy ☐
- I am a bit worried, sad or unhappy ☐
- I am very worried, sad or unhappy ☐

Note: Making any EQ-5D (sample) version available on a publicly accessible webpage is **not allowed**. For reproduction / displaying any EQ-5D sample version, please submit a request for permission by using the EQ-5D registration form.

Table: Recommended age range of users of the EQ-5D versions

Age range	Recommendation
0-3 years	No EQ-5D version is available for this age range.
4-7 years	For children aged 4-7, a proxy version should be used. No self reported EQ-5D-Y is available for this age range at present.
8-11 years	Use EQ-5D-Y. The EQ-5D-Y is more understandable for children in this age range than an adult version of the EQ-5D.
12-15 years	Both the EQ-5D-Y and adult EQ-5D versions can be used. An overlapping area. Generally, EQ-5D-Y is recommended. Depending on the study design, however, it might be appropriate to use one of EQ-5D adult versions. For example, if a study includes both adult respondents and respondents between the age of 12 and 15, the study team might prefer to use just one version of EQ-5D across the whole population.
16 years and over	Use an adult version (EQ-5D-3L or EQ-5D-5L). A possible exception would be a study that only includes children up to age 18. In this case, EQ-5D-Y would be recommended across the full age range, to avoid using two different versions of EQ-5D.

EQ VAS

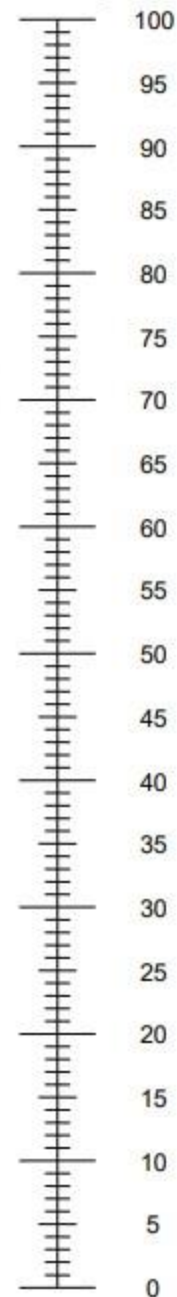
The second part of the questionnaire in all three versions of EQ-5D comprises a standard vertical 20-cm VAS that is calibrated from ‘the worst health you can imagine’ (scored 0) at its base to ‘the best health you can imagine’ (scored 100) at its apex. Respondents are asked to ‘mark an X on the scale to indicate how your health is TODAY’ and to write the number in an adjoining box.

This procedure makes it possible for respondents to provide a rating of their overall health on the day they complete the questionnaire. The VAS can also be used to assess changes in a patient's perception of their own health over time. For example, a patient may rate their current health as 50 on the scale, but after a medical intervention the respondent's rating may be 85, reflecting a substantial improvement in health status. When a group of patients completes the EQ-5D questionnaire before and after treatment, the change in self-rated health can be recorded from the VAS data.

- We would like to know how good or bad your health is TODAY.
- This scale is numbered from 0 to 100.
- 100 means the best health you can imagine.
0 means the worst health you can imagine.
- Please mark an X on the scale to indicate how your health is TODAY.
- Now, write the number you marked on the scale in the box below.

YOUR HEALTH TODAY =

The best health
you can imagine



The worst health
you can imagine

5. Pros and Cons

As we discussed earlier, the EQ-5D (EuroQol-5 Dimensions) is a widely used generic instrument for measuring health-related quality of life which consists of a descriptive system and a visual analog scale (VAS) that assesses overall health status. Here are some pros and cons of using the EQ-5D:

Pros:

1. **Simplicity:** The EQ-5D is relatively simple and easy to administer. It consists of only five dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), making it quick to complete.
2. **Widely used:** The EQ-5D is one of the most commonly used health-related quality of life measures globally. Its widespread use allows for comparisons across different populations and conditions, making it useful for research and policy purposes.
3. **Standardized scoring:** The EQ-5D has a standardized scoring system that assigns utility values to health states based on population preferences. These values facilitate comparisons of health outcomes across different conditions and interventions.
4. **Cost-utility analysis:** The EQ-5D is frequently used in cost-utility analysis to assess the impact of healthcare interventions. Its utility values can be combined with cost data to calculate quality-adjusted life years (QALYs), an important metric in health economics.
5. **Cultural adaptation:** The EQ-5D has been translated into numerous languages and culturally adapted for different populations, allowing for cross-cultural comparisons and ensuring its relevance in diverse settings.

Cons:

1. **Limited dimensions:** The EQ-5D captures only limited dimensions of health-related quality of life, which may not cover all aspects of well-being. It may not capture the nuances of certain conditions or adequately assess specific health domains.
2. **Lack of sensitivity:** The EQ-5D's simplicity can also be a limitation. It may not be sensitive enough to detect small but clinically meaningful changes in health status, particularly for individuals with more complex or severe conditions.
3. **Ceiling and floor effects:** The EQ-5D's scoring system may have ceiling and floor effects, meaning it may not be able to differentiate between individuals with very high or very low levels of health-related quality of life. This can limit its ability to capture changes in these extreme ranges.

4. Preference-based scoring: The EQ-5D utility values are based on population preferences, which may not align with individual preferences or values. The use of generic preference weights may overlook condition-specific aspects that are important to patients.
5. Lack of disease-specific information: The EQ-5D does not provide disease-specific information or detailed insights into specific health conditions. For a more comprehensive assessment, disease-specific measures or additional instruments may be necessary.

It's important to note that while the EQ-5D has its limitations, it remains a widely used and valuable tool for measuring health-related quality of life, especially in population health research and economic evaluations. Its simplicity and standardization make it suitable for large-scale studies and cross-comparisons, but it may not capture the full breadth of an individual's health experience.